

Shashank Mankala

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Summary

AI Engineer / Data Scientist specializing in multimodal ML, time-series forecasting, and production-grade LLM systems. Experienced in building end-to-end pipelines spanning computer vision, embedding-based retrieval, transformer architectures, and scalable MLOps across cloud environments.

Experience

Data Scientist Intern | Tiger Analytics | New York, United States | February 2026 - Present

- Led ML/AI development of an AI-powered garment cost prediction system for a **Fortune 500 retailer**, estimating production costs directly from designer sketches using multimodal embeddings and similarity retrieval
- Built a multimodal embedding pipeline using **FashionCLIP** and **Gemini multimodal embeddings**, combined with KNN-based similarity search to match new designs against historical style cost profiles
- Developed a **Temporal Fusion Transformer (TFT)** model for multi-horizon fabric cost forecasting, capturing temporal trends and price dynamics in raw material markets

Data Scientist Intern | New Era Cap | Buffalo, United States | September 2025 - December 2025

- Designed and deployed an end-to-end demand forecasting system with automated retraining pipelines, integrating **Snowflake and Azure** for scalable batch inference and data orchestration
- Executed model-driven allocation logic and capacity planning workflows, productionizing forecasting outputs for downstream systems across **40k+ SKUs**
- Partnered with engineering teams to define data schemas, build ETL pipelines, validate model outputs, and ensure reproducibility, monitoring, and auditability of ML workflows

Lead Data Analyst | Shadowfax | Bangalore, India | September 2023 - July 2024

- Engineered scalable data pipelines and analytical workflows to optimize logistics routing and workforce allocation, directly informing high-volume operational decision-making
- Collaborated with cross-functional stakeholders to translate complex requirements into enhanced performance monitoring frameworks and data-backed system improvements

Projects

Production-Grade Video Summarization and Q&A Tool | [LINK](#) 

- Designed and deployed a distributed, asynchronous video processing pipeline using FastAPI, Redis, and RQ workers to handle long-form YouTube videos with background ingestion, job tracking, and fault recovery
- Deployed the system on cloud infrastructure using Docker Compose, addressing real-world challenges such as platform bot detection, CPU-only ML constraints, and long-running background workloads

High-Performance LLM Inference Engine with Agentic AI | [LINK](#) 

- Built a local LLM inference engine with full logits-to-decoding pipeline, supporting greedy and probabilistic decoding using temperature, top-k/top-p, and grammar-constrained text generation through token masking and incremental parsing
- Designed an OpenAI-compatible Chat Completions API with streaming, Jinja2-based prompt templates, and inference metrics (TTFT, tokens/sec), benchmarking Hugging Face and vLLM backends to evaluate latency and memory tradeoffs

Education

State University of New York at Buffalo

Master's in Data Science

August 2024 - December 2025

Buffalo, United States

Lovely Professional University

Bachelor's in Computer Science and Engineering

August 2019 - May 2023

Punjab, India

Skills

Languages: Python, SQL, R, C++, JavaScript

Machine Learning and AI: FashionCLIP, Gemini Multimodal Embeddings, KNN Similarity Search, Temporal Fusion Transformer, Computer Vision, LLMs, Transformer Architectures, Decoding Strategies, RAG Systems, Natural Language Processing, Embeddings, Vector Databases, Model Evaluation, PyTorch, TensorFlow, LangChain

MLOps & Data Engineering: Docker, Kubernetes, MLflow, CI/CD, Airflow, Spark, Distributed Systems, Model Deployment, Inference Optimization, Version Control, Git, APIs, Scalable Automation

Cloud Platforms: AWS, Google Cloud (GCP), Azure, Snowflake, Databricks

Data Visualization: Tableau, Power BI, Looker Studio

Database Systems: MySQL, PostgreSQL, MongoDB, BigQuery, FAISS, Redis